Associative Memory Manifolds: A Geometric Integration of Perception and Memory

Anonymous CogSci submission

Abstract

In this paper, we propose the Associative Memory Manifold (AMM), a geometric framework for modeling memory as a dynamic latent space exhibiting human-like memory properties. Unlike discrete attractor networks or address-based architectures, the AMM operates on a continuous latent manifold where information storage is realized through local topological deformations that create basins of attraction. Retrieval is framed as a continuous-time dynamical system, where inputs undergo relaxation dynamics across the manifold surface until reaching a stable equilibrium. Forgetting is defined as a global elastic relaxation of the manifold toward its undeformed state. The framework can be extended to support multi-modal representation. AMM naturally reproduces key cognitive phenomena consistent with human memory results, including the Ebbinghaus forgetting curve, interference effects, and the spacing effect. These results demonstrate that complex memory dynamics, including retention limitations, retrieval latency, and interference effects, emerge intrinsically from the geometry of a continuous, deformable representational space.

Keywords: associative memory; latent space; perception; manifold; attractor dynamics; cognitive efficiency; dynamical system

Introduction

Human memory demonstrates remarkable capabilities alongside apparent limitations. It is capable of robust pattern completion from partial cues, multi-modal integration of visual, acoustic, and abstract information. In this paper, we propose the Associative Memory Manifold (AMM), a geometric framework that integrates perception and memory in a shared latent space. Storage deforms the manifold locally to create attractor basins; retrieval evolves as a dynamical system following energy minimizing trajectories; and forgetting restores flatness globally. Unlike prior models such as the classical discrete Hopfield Network (Hopfield, 1982) or the address-based architecture of Kanerva's Sparse Distributed Memory (Kanerva, 1988), our model utilizes a continuous deformable manifold in a shared latent space, allowing for a continuous representation of memory and a capacity for multi-modal representation (Barua et al., 2023) that naturally demonstrates human-like memory efficiencies and inefficiencies. The model successfully replicates classical memory phenomena such as: forgetting curve (Ebbinghaus, 1913) and interference effects (Underwood, 1957). In this way, we provide an integrated representation of perception and memory that provides natural mechanisms for different memory operations consistent with human-like memory results.

Background

Hopfield Networks

Hopfield Networks (Hopfield, 1982) utilize an energy minimization framework to retrieve stored memories from partial or noisy inputs. In these networks, memories are represented as fixed-point attractors in a discrete state space $S \in \{-1, 1\}^n$, with energy E defined by Hopfield, 1982 as:

$$E = -\frac{1}{2}\sum_{i,j} w_{ij} s_i s_j \tag{1}$$

where w_{ij} represents the symmetric synaptic weight between neuron i and neuron j, and s_i, s_j denote the states of the respective neurons.

While classical Hopfield networks suffer from limited capacity, recent advancements in Dense Associative Memories (Krotov & Hopfield, 2016) and Modern Hopfield Networks (Ramsauer et al., 2021) have introduced continuous energy landscapes and interaction functions that dramatically increase storage capacity. However, while these modern architectures focus primarily on computational capacity and attention mechanisms in transformers, the Associative Memory Manifold (AMM) focuses on the geometric integration of these dynamics within a variational latent space to specifically model human cognitive phenomena.

Sparse Distributed Memory

Our framework draws inspiration from Kanerva's Sparse Distributed Memory (SDM; Kanerva, 1988), which posits that the brain handles high-dimensional sensory data by mapping it onto a vastly larger physical address space where only a fraction of "hard locations" are active. While SDM relies on Hamming distances between discrete bit vectors, the Associative Memory Manifold (AMM) generalizes this to a continuous geometry. In AMM, the "addresses" are points in the latent space, and the storage mechanism replaces hard locations with overlapping Gaussian kernels. This transition from binary to continuous space allows for smoother interpolation and more nuanced similarity metrics than the rigid spheres of activation in traditional SDM.

Hebbian Learning

The deformation of the manifold serves as a geometric manifestation of Hebbian principles (Hebb, 1949). Classically, Hebbian learning is implemented implemented via weight updates between discrete nodes:

$$\Delta w_{ij} \propto x_i x_j$$

Where:

- Δw_{ij}: The change in synaptic weight between the pre-synaptic node j and the post-synaptic node i.
- x_i: The activation level of the post-synaptic node i.
- x_j: The activation level of the pre-synaptic node j.

In AMM, this strengthening is represented by deepening of the potential well. When an experience is repeated, the depth of corresponding memory increases, effectively lowering the energy barrier for retrieval. This unifies the "neurons that fire together, wire together" rule with the topology of the latent space, where frequent co-activation creates a steeper gradient, ensuring that the system naturally gravitates toward frequently reinforced patterns.

From Discrete Attractors to Continuous Manifolds

The transition from Hopfield dynamics to the Associative Memory Manifold (AMM) represents a shift toward a topologically adaptive latent surface. While classical models are limited to discrete hypercubes (Hopfield, 1982), Modern Hopfield Networks (Ramsauer et al., 2021) introduce continuous states but prioritize storage capacity via sharp energy wells. This often results in 'all-or-nothing' retrieval dynamics.

In contrast, the AMM defines a potential field $V(z)$ over a continuous manifold $Z \in \mathbb{R}^d$ designed to mirror human memory constraints. Storage is achieved by deforming the manifold's topology. This formulation allows for content-addressable retrieval via gradient flow. Because this landscape preserves the geometric relationships between memories, the model enables a gradient of recall accuracy that mirrors human reaction times and confidence levels.

Methodology

The framework maps each sensory input to a manifold based on the embedding produced by the encoder. Let X be the experiences encoded as a vector, then an experience x_i will be mapped to a location z_i in the manifold Z.

s_i: Strength of the i-th stored memory
σ: Radius of effect
μ_i: Latent position of the i-th stored memory
κ: Elasticity
η: Rate of descent
z: Latent position of the current experience

Storage

AMM's encoder maps each input x to an embedding z. Each experience of x will form a deformation ("dent") around z based on the strength of the experience. This deformation forms an attractor basin. The depth and width of the curvature reflect the experience's salience, frequency, and recency. The total potential V at any point z is the sum of all stored experiences:

$$V(z) = \sum_i -s_i \exp\left(-\frac{\|z - \mu_i\|^2}{2\sigma^2}\right) \tag{2}$$

Retrieval

Retrieval is based on gradient descent. The input x will be mapped to z. Based on the existing deformations around z, we roll down the attractor basin until we reach a stable point z_i which is the strongest previous experience. In this way, retrieval is guided by previous experiences entirely based on the geometry of Z. The following equation describes the step-by-step movement of a cue z towards a stable memory basin.

$$z_{t+1} = z_t - \eta \nabla V(z_t) \tag{3}$$

The speed and success of retrieval are governed by two main factors:

- Basin Depth (Strength): For highly salient or frequently reinforced memories, the potential well is deep with steep gradient. This will cause the cure to accelerate and quickly reach the stable point.
- Descent Duration: If retrieval is interrupted or limited by a time constraint, the cue may only reach an intermediate point on the manifold retrieving a fuzzy memory around the strong memory basin. This accurately models human memory performance where rapid retrieval is less accurate and is prone to interference from nearby attractors.

By treating the number of steps required to reach a convergence as a proxy for reaction time and exploration time, the model replicated key memory properties:

- Strong memories will be quickly retrieved with high accuracy as the cure settles at the stable point with fewer steps
- Weak memories will require more iterations for the cue to settle at a stable point, demonstrating a longer retrieval time when humans struggle to recall faint memories

Forgetting

Forgetting is a property of the manifold itself. Each experience forms a deformation in the manifold and the manifold tends to change back to its original undeformed state. The manifold naturally tends toward its minimal energy state (flatness). We model this relaxation as a global restorative force proportional to the current depth of the attractor:

$$\frac{ds_i}{dt} = -\kappa s_i \tag{4}$$

Experiments

We utilized the Associative Memory Manifold to simulate core phenomena of human memory. The model represents memories as attractors in a high-dimensional latent space,

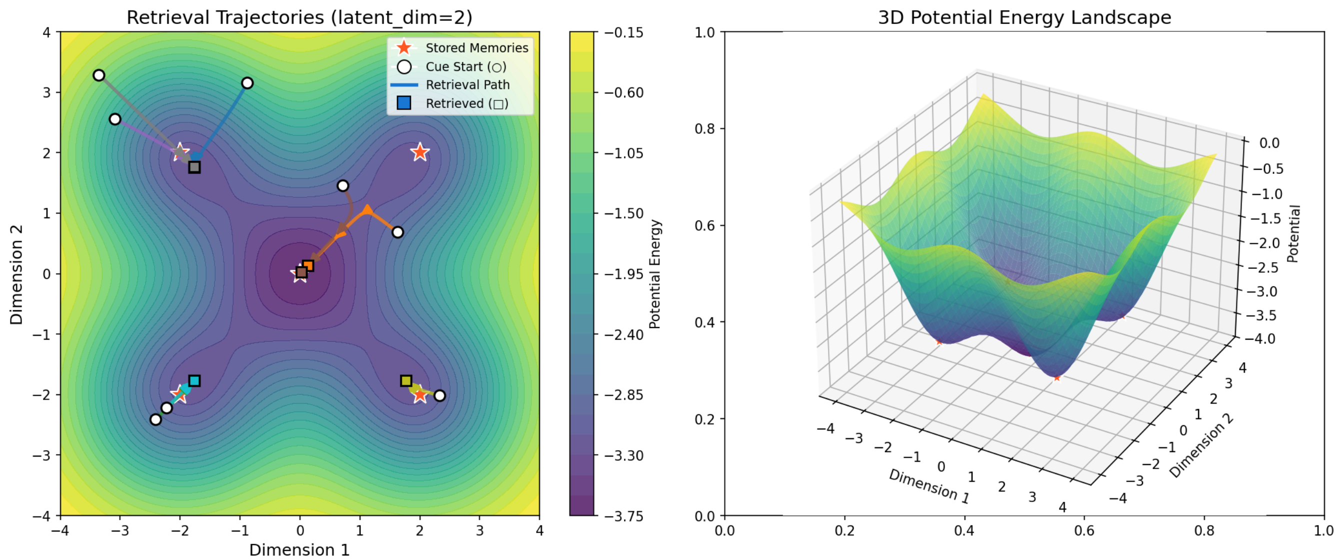

Figure 1: The Associative Memory Manifold (AMM) in a 2D latent space. The left panel illustrates the gradient-based retrieval trajectory from a start cue to a stable memory state. The right panel depicts the surface topology, demonstrating how the manifold is deformed to create stable basins of attraction for each stored memory.

where retrieval is performed via gradient descent on the potential energy landscape defined by the sum of stored memory Gaussians.

Encoder

In the AMM framework, the encoder f_θ is implemented as the inference network of a Convolutional Variational Autoencoder (CVAE; Kingma and Welling, 2014). While a CVAE is used to exploit the spatial relationship, other encoder architectures are viable as well. Unlike a standard deterministic encoder, the VAE maps an input x to a distribution in the latent space Z characterized by a mean μ and a variance σ^2. For memory storage purposes, we utilize the mean vector $\mu(x)$ as the primary coordinate for the attractor basin. The VAE was pre-trained and fixed before the experiments to isolate the geometric memory dynamics from perceptual learning effects. The shared nature of the encoder implies that the act of sensing is inextricably linked to the act of addressing. When a new stimulus is perceived, it is automatically projected onto Z via f_θ. If that stimulus falls within an existing attractor basin formed by prior deformations, retrieval begins immediately within the same representational space. This eliminates the computational overhead of translating between different representational formats and allows for a seamless transition from real-time perception to recall. Consequently, perception is a dynamical process based on the geometric properties of the underlying memory system.

Forgetting Curve

We initialized the manifold with a set of random memories ($N = 5$) and defined the manifold's properties using an elasticity constant $\kappa = 0.1$. The memory strength s over time t was affected by the restorative property of the manifold:

$$s_i(t+1) = s_i(t) \cdot (1 - \kappa), \quad 0 \leq \kappa \leq 1 \tag{5}$$

This discrete update rule corresponds to the solution of the differential equation $\frac{ds}{dt} = -\kappa s$. The simulation measured cumulative memory strength over 50 time intervals. The resulting trajectory naturally reproduces the characteristic Ebbinghaus Forgetting Curve (Ebbinghaus, 1913).

Interference Effects

We investigated how competing memories disrupt retrieval accuracy through proactive and retroactive interference. In both conditions, a target memory (centered at origin) was stored with strength $s_{target} = 2.0$, and retrieval was attempted using a noisy cue.

- **Control (No Interference):** The target was stored in isolation.
- **Proactive Interference:** A confounding "old" memory (similar to the target but distinct) was stored with higher strength ($s_{old} = 3.0$) *before* the target was learned.
- **Retroactive Interference:** A confounding "new" memory was stored with higher strength ($s_{new} = 3.0$) *after* the target was learned.

The stronger, nearby interfering attractors distorted the potential energy landscape, creating local minima that diverted the gradient descent retrieval process (Figure 1). The simulation showed significantly higher retrieval errors (Euclidean distance from target) under both interference conditions compared to the control, mirroring empirical findings in which similar information retrieval is inhibited by competing traces.

Spacing Effect

To evaluate the impact of learning schedules on retention, we compared "Massed Practice" (cramming) versus "Spaced Practice" (distributed learning).

- **Massed Reinforcement:** The target was reinforced 5 times in immediate succession (total strength summed at $t = 0$), followed by a long retention interval of 20 time steps where the manifold was allowed to relax naturally.
- **Spaced Reinforcement:** The target was reinforced 5 times, but with relaxation intervals of 4 time steps inserted between each repetition. The final retrieval test occurred at the same total elapsed time ($t = 20$).

Despite the same number of reinforcements, the Spaced Condition yielded significantly lower retrieval errors and higher final memory strength. In Massed Reinforcement, the single aggregated memory trace was subjected to the manifold's restorative tendency for the full duration ($5 \cdot (1-\kappa)^{20}$). In Spaced Reinforcement, later repetitions occurred closer to the test time and had less time to dissipate ($1 \cdot (1-\kappa)^{20} + \cdots + 1 \cdot (1-\kappa)^{4}$). Consequently, the model captures the 'spacing effect' (Melton, 1970) as an emergent geometric property.

Results

The simulation results from the Associative Memory Manifold (AMM) framework demonstrate a high degree of qualitative and quantitative alignment with established human memory phenomena.

Temporal Relaxation and Retention

As shown in Figure 2, the model's global relaxation mechanism successfully replicates the Ebbinghaus Forgetting Curve. The total memory strength follows an exponential decay rate, where the rate of information loss is highest immediately following the storage event. This mirrors the findings of Ebbinghaus (1913), who observed that the "savings" in relearning nonsense syllables dropped most sharply within the first 20 minutes and stabilized after 31 days. In our manifold, this corresponds to the rapid relaxation of the potential well $V(z)$ toward a flat state.

Interference and Retrieval Error

Retrieval accuracy was significantly modulated by the presence of competing memory traces (Figure 4).

- **Proactive Interference:** Storing a high-strength attractor near the target prior to learning resulted in a mean retrieval error of 0.372. This simulates the "competition" effect described by Underwood (1957), where older memories inhibit the recall of new information.
- **Retroactive Interference:** A similar error magnitude (0.332) was observed when new memories were introduced after the target.

The box plots in Figure 4 (right) reveal that interference does not just decrease accuracy; it increases the *variance* of retrieval, suggesting that the "cue" is frequently captured by the wrong attractor basin, a phenomenon akin to "false memories" or "blending" in human recall.

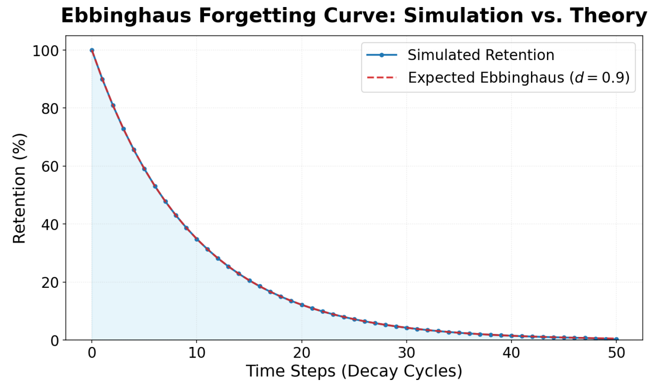

Figure 2: Strength of stored memory over time compared to Ebbinghaus curve

The Spacing Effect

The most robust finding was the advantage of distributed practice. Figure 3 compares Massed vs. Spaced practice. While both conditions performed well in the *Immediate Test*, the Massed Condition showed an increase in retrieval error during the *Delayed Test* (≈ 0.16). In contrast, the Spaced Condition maintained a low error rate (≈ 0.05).

This replicates the "Lag Effect" identified by Melton (1970), where increasing the interval between study sessions enhances long-term retention. In the AMM framework, spacing allows the manifold to settle between reinforcements, effectively "stacking" the potential wells in a way that resists global decay more effectively than a single, high-intensity deformation.

Discussion

The Associative Memory Manifold (AMM) provides a structural explanation for why human memory is simultaneously remarkably robust and frustratingly fallible. By having the perception and memory work within a shared latent space, the system prioritizes representational efficiency over informational fidelity. This trade-off aligns with the Principle of Efficient Coding (Barlow, 1961), suggesting that the "inefficiencies" of biological memory are emergent properties of a geometry optimized for survival rather than perfect recall.

Forgetting as Manifold Regularization

In traditional computational systems, forgetting is often treated as a failure of storage or a failure of retrieval. In the AMM, forgetting (manifold relaxation) acts as an adaptive regularization mechanism. By allowing high-frequency, low-salience deformations to flatten over time, the system prevents the latent space from overfitting to ephemeral noise. This global relaxation ensures that only statistically significant or emotionally salient patterns maintain deep attractor basins, maintaining the manifold's "smoothness" and ensuring the encoder remains generalizable to new experiences.

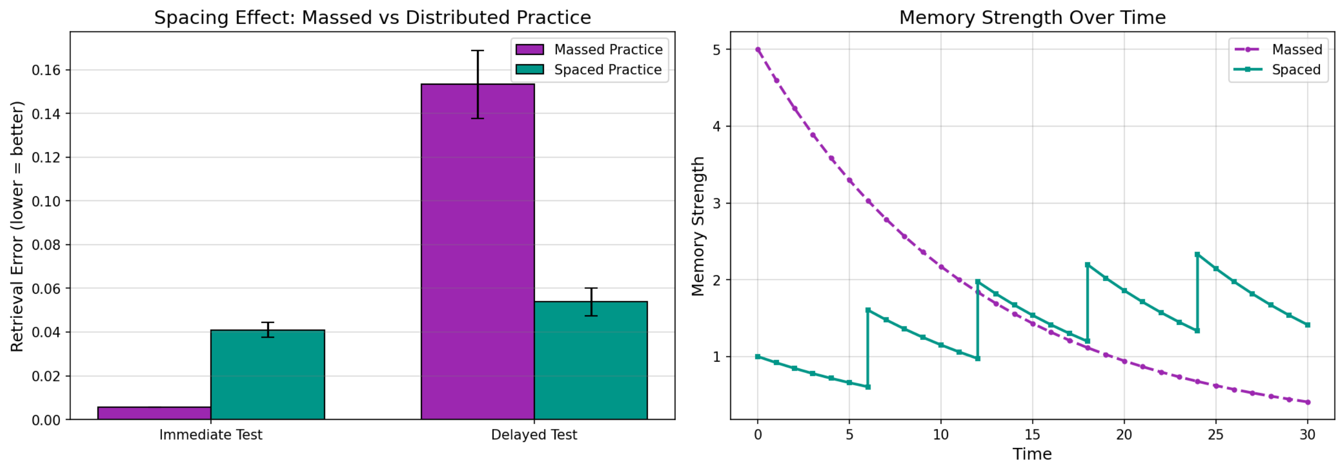

Figure 3: Memory Retrieval Accuracy (Massed Experiences vs Spaced Experiences)

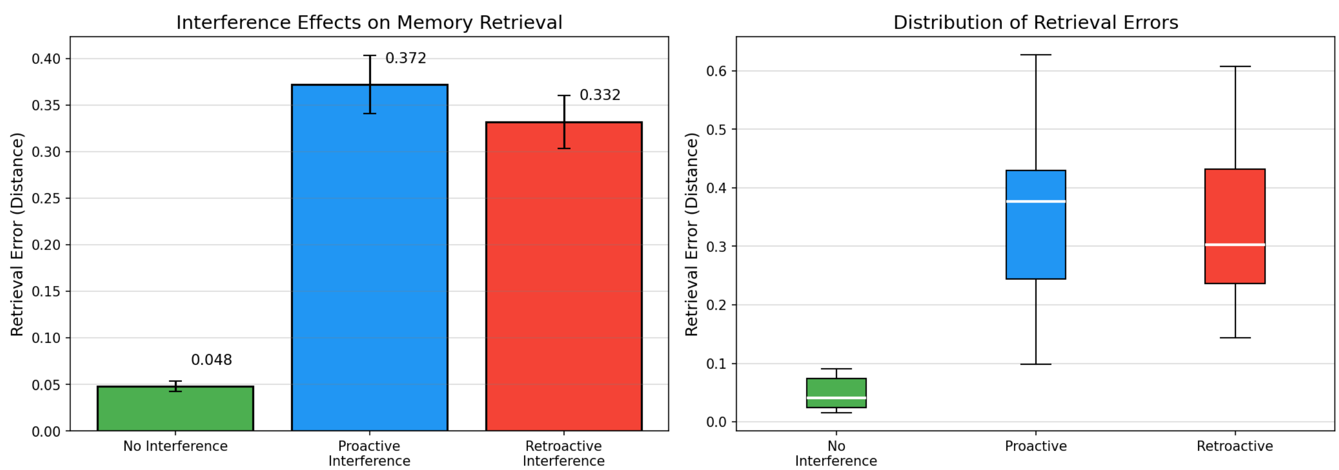

Figure 4: Retrieval accuracy with interference from other memories

Interference as a Byproduct of Generalization

The interference effects we observed demonstrate that retrieval errors occur when two distinct experiences are mapped to proximal coordinates in Z. While this causes "blending" or false memories, it is the fundamental mechanism behind knowledge generalization. The same geometric proximity that causes interference also allows the system to project the learned dynamics of prior experiences onto new, similar stimuli.

Temporal Inefficiency and Reaction Times

The AMM treats retrieval as a physical journey through a potential landscape. The "inefficiency" of slow recall for weak memories is represented by the lower gradient $\nabla V(z)$ in shallow basins, requiring more iterations to reach convergence. Rather than an architectural flaw, this temporal delay serves as a confidence metric: the system only "commits" to a memory quickly when the geometric evidence (the depth of the basin) is overwhelming.

Conclusion

In this work, we introduced the Associative Memory Manifold (AMM), a geometric framework that defines as a continuous dynamical evolution governed by manifold geometry. While Modern Hopfield Networks rely on discrete point attractors and Sparse Distributed Memory (SDM) depends on fixed address radii, AMM embeds information into a continuous manifold. This allows retrieval to unfold as smooth trajectories. By eliminating the rigid discretizations found in classical SDM and the isolated basins of Hopfield models, AMM enables a seamless transition from real-time perception to recall avoiding the overhead of translation between internal formats. Consequently, AMM offers a robust, scalable architecture that generalizes the principles of distributed storage into a fully continuous dynamical domain.

By utilizing deformations on the latent space for storage and gradient descent for retrieval, the AMM provides a content-addressable system that handles noisy and partial cues with biological plausibility. This integration offers a computationally efficient alternative to modular architectures.

Future Work

While the AMM provides a robust foundation for a geometric integration of perception and memory, several critical avenues for expansion remain to fully capture the complexity of biological cognition:

- **Representational Drift and Manifold Stability:** Since memories are stored as topological "dents" tied to specific latent coordinates, updating the underlying encoder introduces the risk of representational drift. Future work must address the stability-plasticity dilemma: finding mechanisms to update the latent space Z to learn new features without fracturing the existing geometry or "misplacing" previously stored attractor basins.
- **Conscious Control over Memory Exploration:** In its current form, retrieval in AMM is a dynamical process dictated strictly by the manifold's local gradient and the time duration of the descent. However, human retrieval often involves top-down, volitional modulation. We aim to investigate how a "task-dependent" bias vector could warp the potential landscape $V(z)$ in real-time, allowing the system to exert conscious control over the direction of memory search and exploration.
- **Mechanisms for Storing Sequences of Data:** Human memory is inherently episodic and sequential. AMM currently doesn't have the mechanism to store temporal sequences where the completion of one retrieval event naturally triggers the gradient descent toward the next chronological memory state.

Acknowledgments

AI-driven auto-complete and code-generation tools were employed during the implementation of the simulation environment.

References

Barlow, H. B. (1961). Possible principles underlying the transformation of sensory messages. In W. A. Rosenblith (Ed.), *Sensory communication* (pp. 217–234). MIT Press.

Barua, A., Ahmed, M. U., & Begum, S. (2023). A systematic literature review on multimodal machine learning: Applications, challenges, gaps and future directions. *IEEE Access*, *11*, 14804–14831. https://doi.org/10.1109/ACCESS.2023.3243854

Ebbinghaus, H. (1913). *Memory: A contribution to experimental psychology* (H. A. Ruger & C. E. Bussenius, Trans.) [Original work published 1885]. Teachers College, Columbia University.

Hebb, D. O. (1949). *The organization of behavior: A neuropsychological theory*. Wiley.

Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, *79*(8), 2554–2558.

Kanerva, P. (1988). *Sparse distributed memory*. MIT Press.

Kingma, D. P., & Welling, M. (2014). Auto-encoding variational bayes. *International Conference on Learning Representations (ICLR)*.

Krotov, D., & Hopfield, J. J. (2016). Dense associative memory for pattern recognition. *Advances in Neural Information Processing Systems*, *29*. https://proceedings.neurips.cc/paper/2016/hash/eaae339c4d89fc102edd9dbdb6a28915-Abstract.html

Melton, A. W. (1970). The situation with respect to the spacing of repetitions and memory. *Journal of Verbal Learning and Verbal Behavior*, *9*(5), 596–606.

Ramsauer, H., Schäfl, B., Lehner, J., Seidl, P., Widrich, M., Adler, T., Gruber, L., Holzleitner, M., Pavlović, M., Sandve, G. K., Greiffenstein, V., Kreil, D., Kopp, M., Klambauer, G., Brandstetter, J., & Hochreiter, S. (2021). Hopfield networks is all you need. *International Conference on Learning Representations*. https://openreview.net/forum?id=tBMl04R74L

Underwood, B. J. (1957). Interference and forgetting. *Psychological Review*, *64*(1), 49–60.